

Housing Price Analysis — Summary

Dataset

The dataset contains 545 properties with 13 columns and no missing values. The target variable is price, predicted from 12 features covering size and amenities: area, bedrooms, bathrooms, stories, parking, and binary indicators for mainroad access, guestroom, basement, hot water heating, air conditioning, preferred area, and furnishing status.

Methodology

Duplicate rows were removed and binary yes/no fields were converted to 0/1; furnishing status was one-hot encoded. The data was split 80/20 into training and test sets. Two regression models were trained to predict price: Linear Regression and a Random Forest Regressor, evaluated using MAE, RMSE, and R^2 on the held-out test set.

Model Performance

Model	MAE	RMSE	R^2
Linear Regression	970,043	1,324,507	0.653
Random Forest	1,022,560	1,401,497	0.611

Linear Regression slightly outperformed Random Forest on this dataset, suggesting the relationship between features and price is largely linear and that Random Forest's added flexibility did not translate into better generalization at this sample size.

Key Insights

Area is the strongest driver of price, followed by the number of bedrooms and bathrooms. Being in a preferred area and having more parking spaces are both associated with higher prices, while furnishing status has a smaller effect than expected. Prediction errors remain large relative to absolute prices, so model output is best used as directional guidance rather than precise valuation.

Recommendation

For pricing and marketing, prioritize larger, well-located properties and lead with quantifiable attributes (area, bedrooms, bathrooms, preferred area) to justify asking price. Collecting additional features such as property age and renovation history would likely improve future model accuracy.